Install and run OpenDrift

OpenDrift documentation is available here:
https://opendrift.github.io/

Installation is already done at the desktops in dataroom U327. The students must create a conda environment and install OpenDrift on their own user profile as described in the presentation.

To run the OpenDrift model:

Create a folder where you can store the example scripts and model output. The python scripts (*.py files) can be found and downloaded from https://github.com/magnesim/opendrift_simulations and can be modified with your favourite text editor.

To perform a model simulation, first go into the folder where the examples are stored and activate the mamba/conda environment:

```
$ conda activate opendrift
```

Then modify the basic surface drift example and run the command

```
$ python rad320_surface.py
```

A more advanced example of radionuclide drift can be modified and run:

```
$ python rad320_radionuclides.py
```

An example of postprocessing of the model results can be found here:

```
$ python rad320_trajan_analysis.py
```

Documentation of the Trajan package can be found here: https://opendrift.github.io/trajan/

Exercise 1

Run the generic example of surface drift with a pulse discharge at 58.9N, 10.5E at 2025-10-14 12:00UTC

1. Run 24 hours simulation
2. Plot the simulated trajectories
3. Add sea surface currents to the animation
4. Run the same simulation with a release today at 1200UTC

Exercise 2

Compute the transport of a nuclear discharge of total 1 TBq (10^{12}) Cs-137 to the Oslo Fjord near Drøbak at 2025-10-14 12:00 UTC. The discharge distribution is 50% LMM and 50% particle-bound species.

1. Run a simulation for 48 hours with continuous discharge through the simulation period
2. Compute concentration levels using the rad320_trajan_analysis.py script
3. What are the maximum concentration levels in a 5m thick surface layer? Which locations have the highest exposure?
4. Repeat the simulations, with variations of the configuration: How does this affect the contamination pattern?
    a. Change the discharge to 5% LMM and 95% particle-bound species.
    b. Apply high resolution currents (160m instead of 800m). How does this affect the contamination pattern?
    c. Change the mixing by adjusting the horizontal and vertical diffusion coefficients
    d. Run the same simulation with a release today at 1200UTC
5. Plot time series from selected locations
6. Plot only LMM species / particle species